

# DAVID ARTHURIAN

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## EXPERIENCE

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### Northrop Grumman

May 2025 - Present

#### Electrical Engineer

- Serve as Cognizant Engineer for the Electronic Power Subsystem (EPS) on NASA's HALO spacecraft, leading design, integration, and verification of power distribution hardware supporting critical spaceflight operations.
- Develop, analyze, and validate power electronics and PCB-level hardware, ensuring high-reliability operation in radiation and thermal environments.
- Collaborate with cross-functional teams (systems, mechanical, software, and suppliers) to drive requirements, root-cause hardware issues, and deliver mission-critical hardware under aggressive schedules.
- Conduct circuit-level simulations (LTSpice, OrCAD, Simulink) and hardware bring-up/debug using oscilloscopes, power analyzers, and automated test scripts (Python/Matlab).
- Lead failure analysis and corrective actions for hardware anomalies, improving reliability and reducing integration risk.
- Author and review technical documentation, schematics, and test procedures to meet NASA and industry standards.

### Sempra Energy

June 2023 - May 2025

#### Systems Engineer

- Collaborated with an engineering team to design and optimize Southern California CNG/LNG distribution systems for a large-scale critical infrastructure network serving over 500,000 residential, commercial, and industrial customers.
- Developed real-time monitoring and analytics tools using SCADA, GIS, SAP, and SQL to enhance system performance.
- Created emergency response dashboards for system engineers, ensuring rapid diagnostics and failure mitigation.
- Led instrumentation and control system projects, integrating hardware and software for high-reliability gas distribution.
- Conducted data modeling with SQL Data to implement system improvements to improve CNG distribution by 50%.
- Compiled data, created dashboards for emergency scenarios, and provided on-call support for region engineers.
- Designed and executed six-figure capital projects, achieving 25% cost savings and delivery ahead of schedule.

## PROJECTS

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### Embedded Systems - Video Game

Jan. 2024 - May 2024

- Designed and built an interactive, sensor-based multiplayer video game with IoT communication.
- Created hardware-based motion controllers using Arduinos, IMU sensors, and LEDs for gesture-based input.
- Embedded real-time embedded control and wireless communication protocols for multiplayer synchronization.
- Engineered the game logic in Python and developed a Unity-based graphical interface for stable and fun gameplay.

### Micromouse Competition - 1<sup>st</sup> Place

April 2023 - May 2023

- Programmed and assembled an autonomous robotic vehicle for maze navigation using PID controllers and IR sensors.
- Integrated real-time embedded control algorithms in C++, optimizing performance for speed and accuracy.
- Achieved 1st place out of 30 participants, completing the challenge with a record time of sub 10 seconds.

### Network Attached Storage (NAS)

July 2022

- Produced and built a custom NAS system using Raspberry Pi 4 and RAID-configured HDDs to store personal data.
- Configured network protocols and security layers for secure cloud data storage and remote access by individuals.

## SKILLS

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C++, Python, Asyncio, MATLAB, C++, SQL, Verilog, VHDL, JavaScript, LTSpice, OrCAD, Altium, PCB Design, Simulink, Oscilloscopes, Power Analyzers, IBM DOORS, Cameo, Creo, Windchill, Git, VSCode, SCADA, SAP, GIS

## EDUCATION

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### The University of California, Los Angeles

Sept. 2020 - June 2024

B.S. Electrical Engineering, IEEE Member, HKN Society

GPA: 3.75

EIT Certificate in Electrical Engineering #182187

June 2024